

>	$12 + 4 - 5$	11	(1)
>	2^{10}	1024	(2)
>	$\sin(0.1)$	0.09983341665	(3)
>	$(a + b)(a - b)$	$a(a - b) + b(a - b)$	(4)
>	$f := x \rightarrow 3x^3 + 2x^2 - 5 :$		
>	$\text{diff}(f(x), x)$	$9x^2 + 4x$	(5)
>	$f := x \rightarrow \text{sqrt}(1 + x^4) :$		
>	$\text{diff}(f(x), x)$	$\frac{2x^3}{\sqrt{x^4 + 1}}$	(6)
>	$f := x \rightarrow \exp(x) \cdot \sin(x) \cdot \cos(x) :$		
>	$\text{diff}(f(x), x)$	$e^x \sin(x) \cos(x) + e^x \cos(x)^2 - e^x \sin(x)^2$	(7)
>	$\text{simplify}((a + b)(a - b))$	$a(a - b) + b(a - b)$	(8)
>	$\text{expand}((a + b) \cdot (a - b))$	$a^2 - b^2$	(9)
>	$\text{simplify}((a - b) \cdot (a + b))$	$a^2 - b^2$	(10)
>	$\text{int}(3x^3 + 2x^2 - 5, x = 0 .. 1)$	$-\frac{43}{12}$	(11)
>	$\text{int}\left(\frac{1}{x^2}, x = 0 .. \text{infinity}\right)$	∞	(12)
>	$\text{int}(\exp(-x^2), x = -\text{infinity} .. \text{infinity});$	$\sqrt{\pi}$	(13)
>	$\text{limit}\left(\frac{\sin(x)}{x}, x = 0\right)$	1	(14)
>	$\text{limit}\left(\frac{(x^3 + 3x^2 - 5)}{2x^3 - 7x}, x = \text{infinity}\right)$	$\frac{1}{2}$	(15)

```
> limit( (cos(x) + 1) / (x - Pi), x = Pi)
```

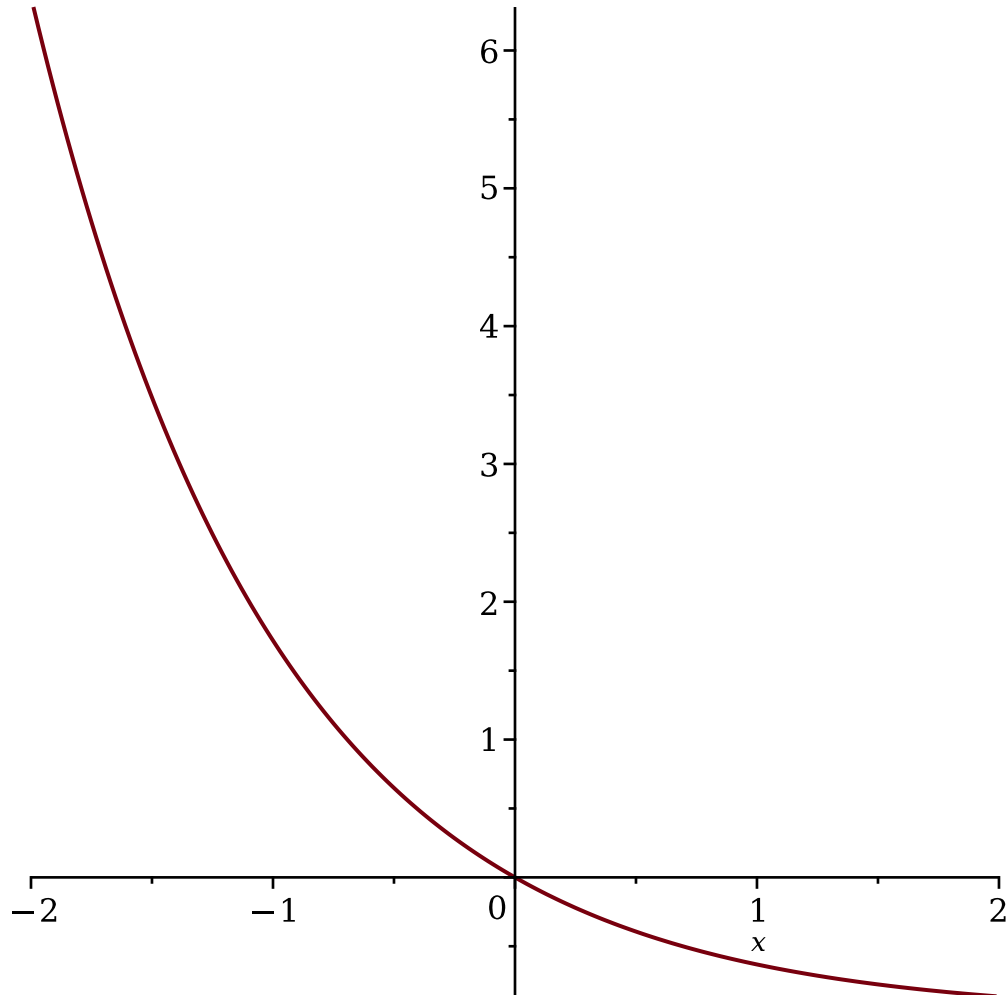
0

(16)

```
> f := x -> exp(-x) - 1 :
```

```
> with(plots) :
```

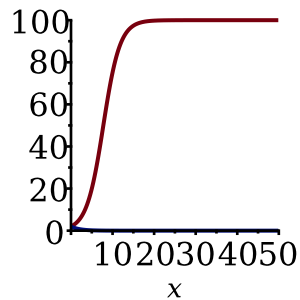
```
> plot(f(x), x = -2..2)
```



```
> f := (x, r) -> (200 * exp(r * x)) / (2 * (exp(r * x) - 1) + 100) :
```

```
> with(plots) :
```

```
> plot([f(x, 0.5), f(x, -0.5)], x = 0..50, color = [red, blue])
```

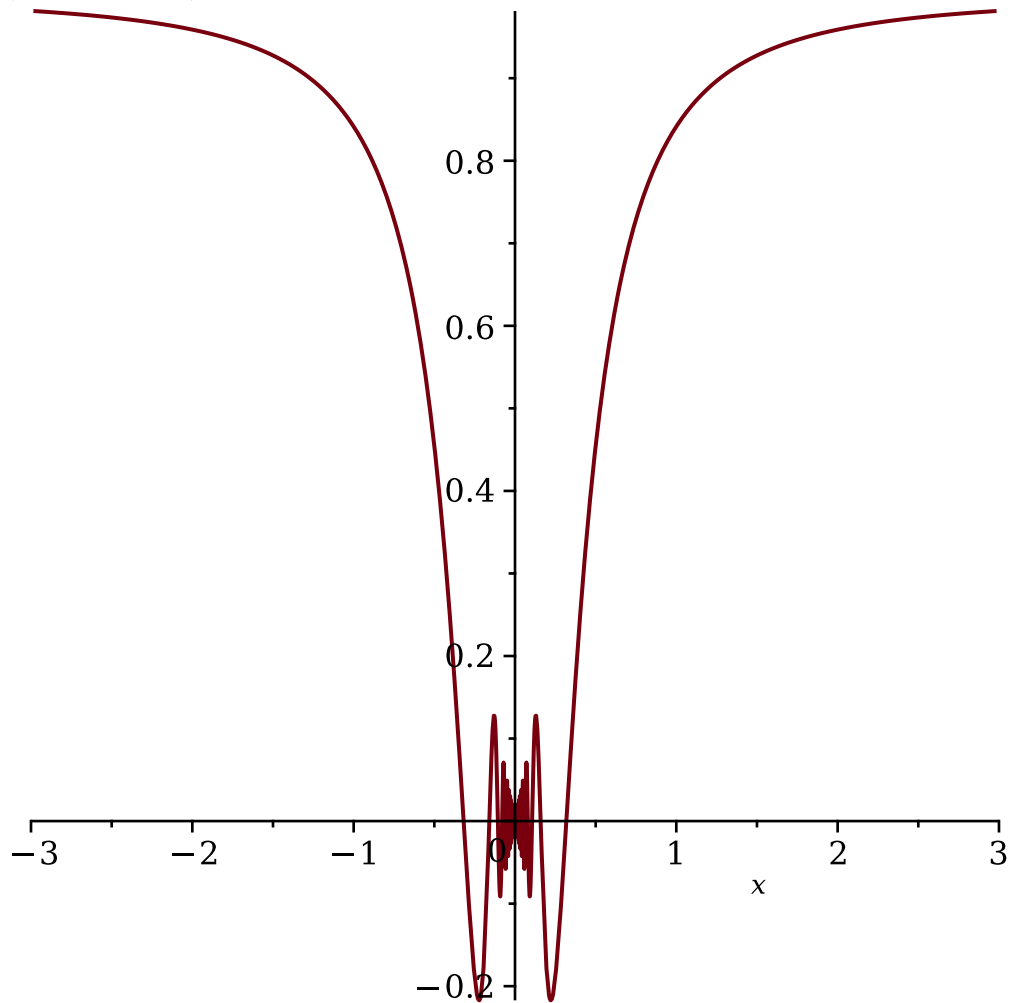


, color = [red, blue]

```
> f := x ↦ x · sin(  $\frac{1}{x}$  ) :
```

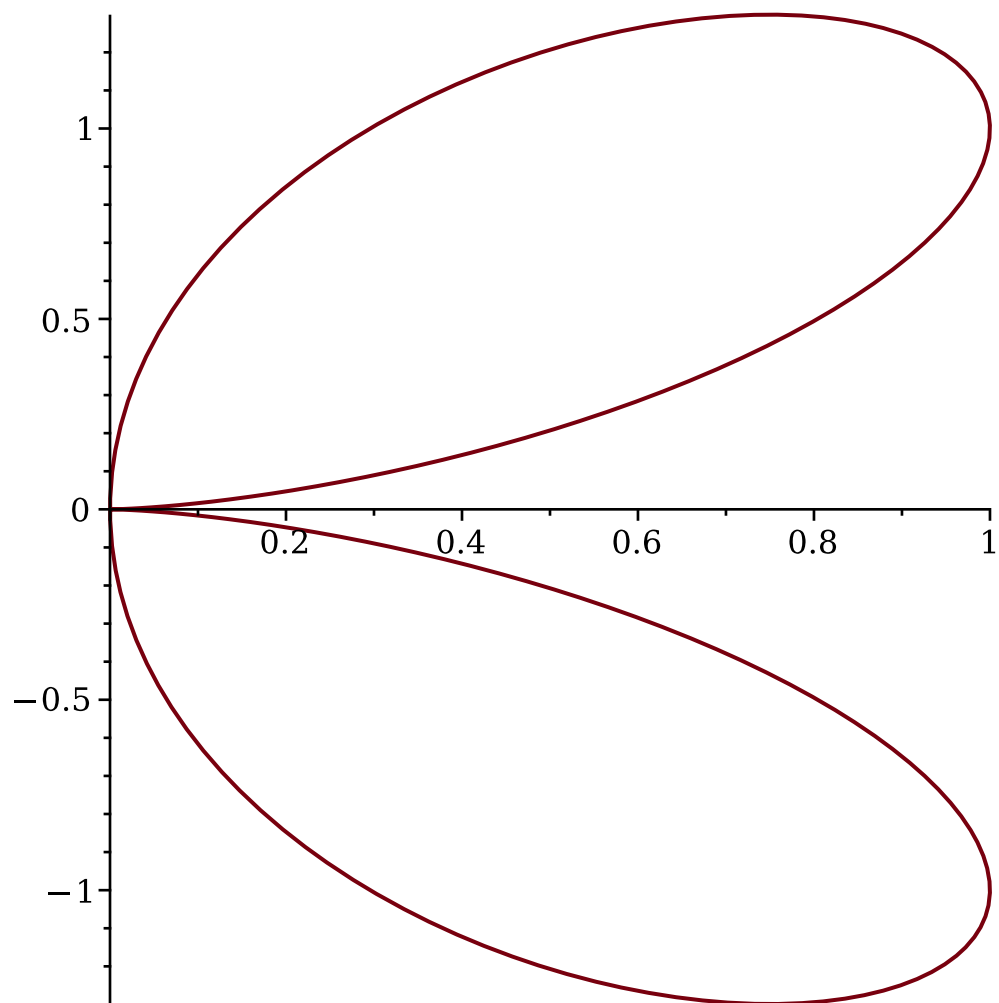
```
> with(plots) :
```

```
> plot(f(x), x = -3..3)
```

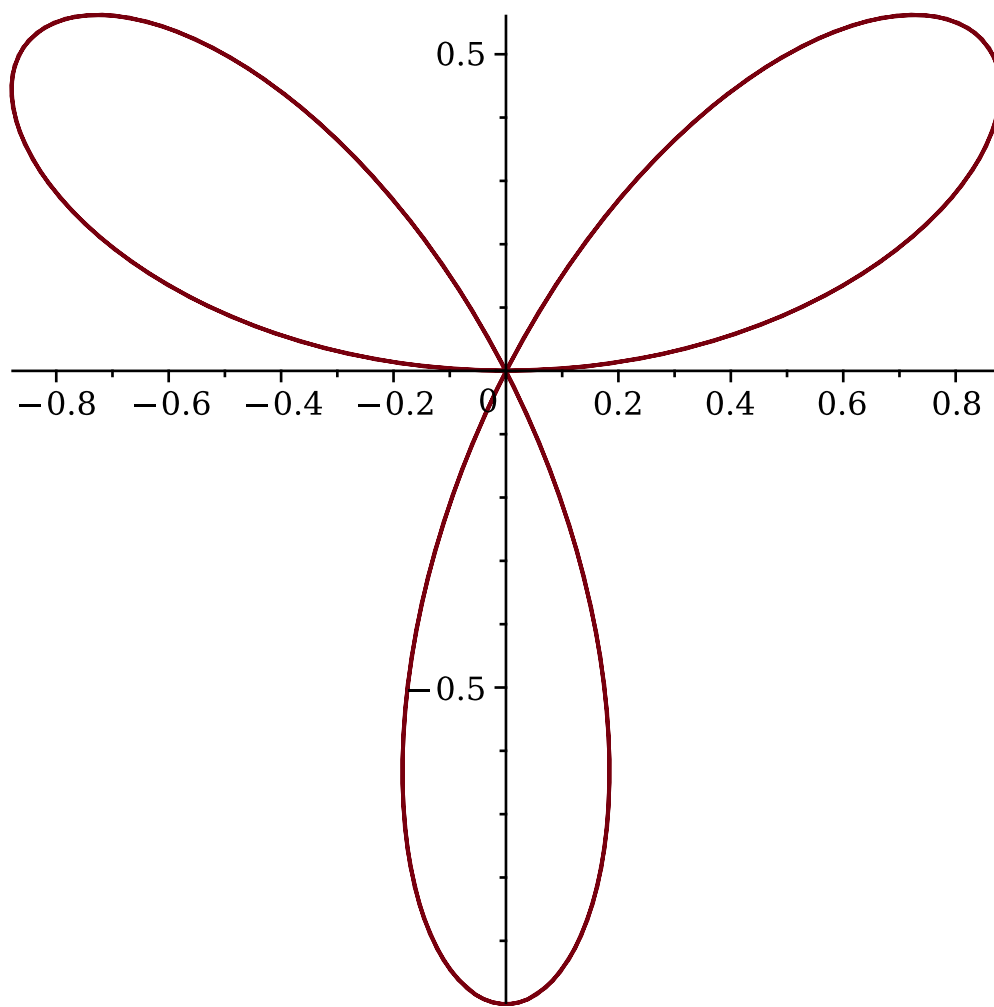


```
> with(plots) :
```

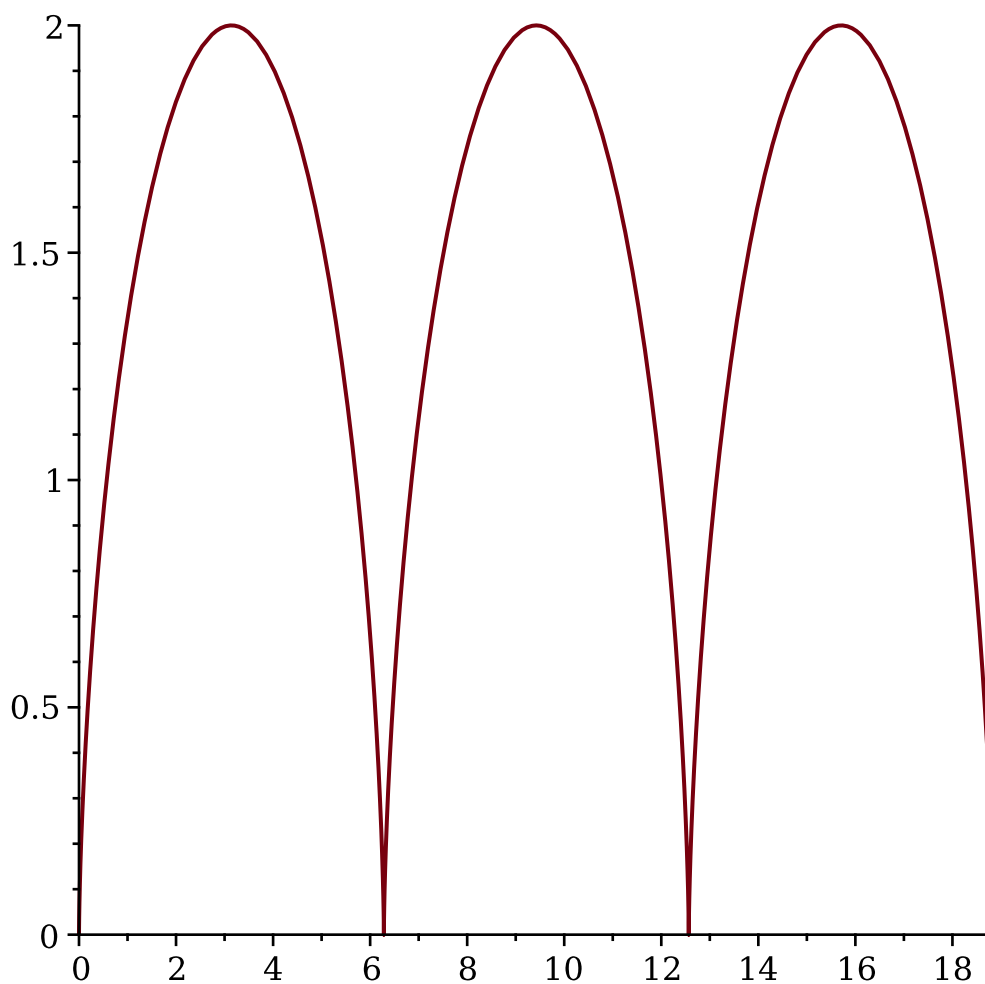
```
> plot([ (1 - cos(t)) · cos(t) , (1 - cos(t)) · sin(t) , t = 0..2 Pi])
```



```
> with(plots) :  
> plot([sin(3·t)cos(t), sin(3·t)sin(t), t = 0..2·Pi])
```



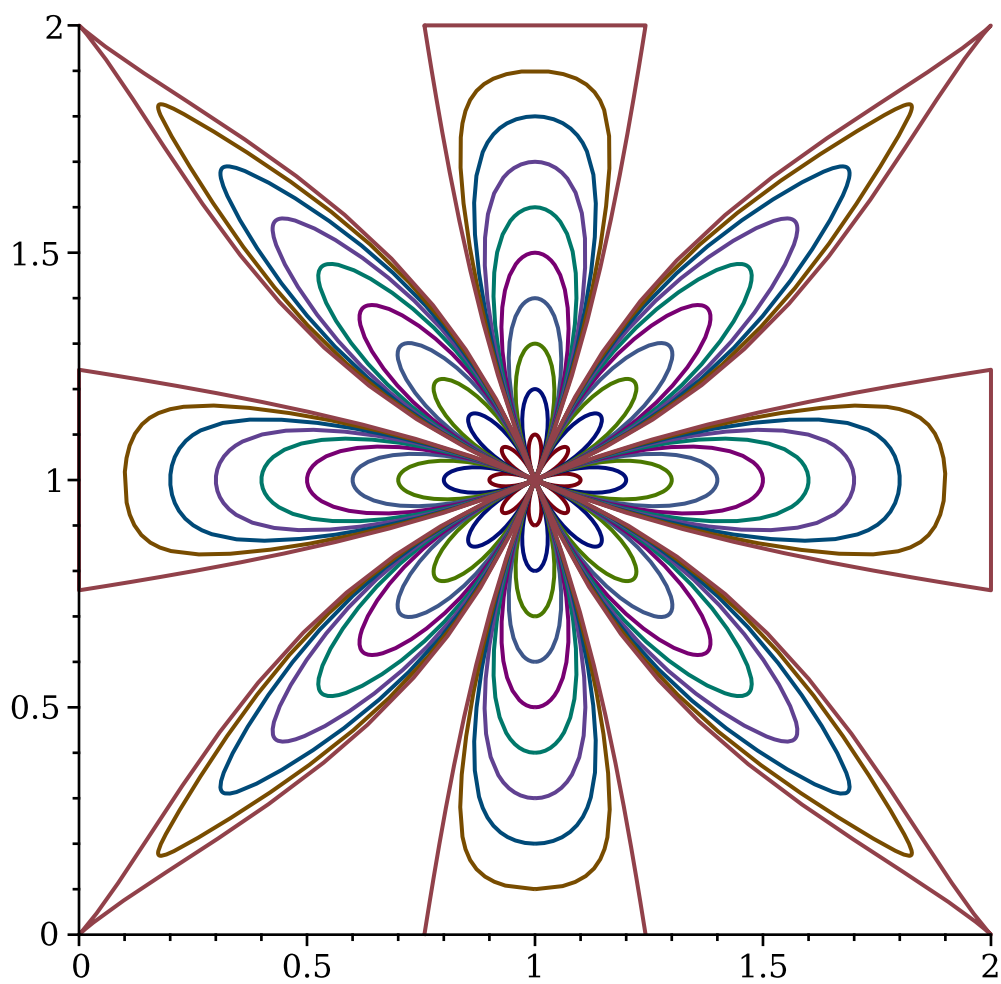
```
> with(plots) :  
> plot([t - sin(t), 1 - cos(t), t = 0..6·Pi])
```



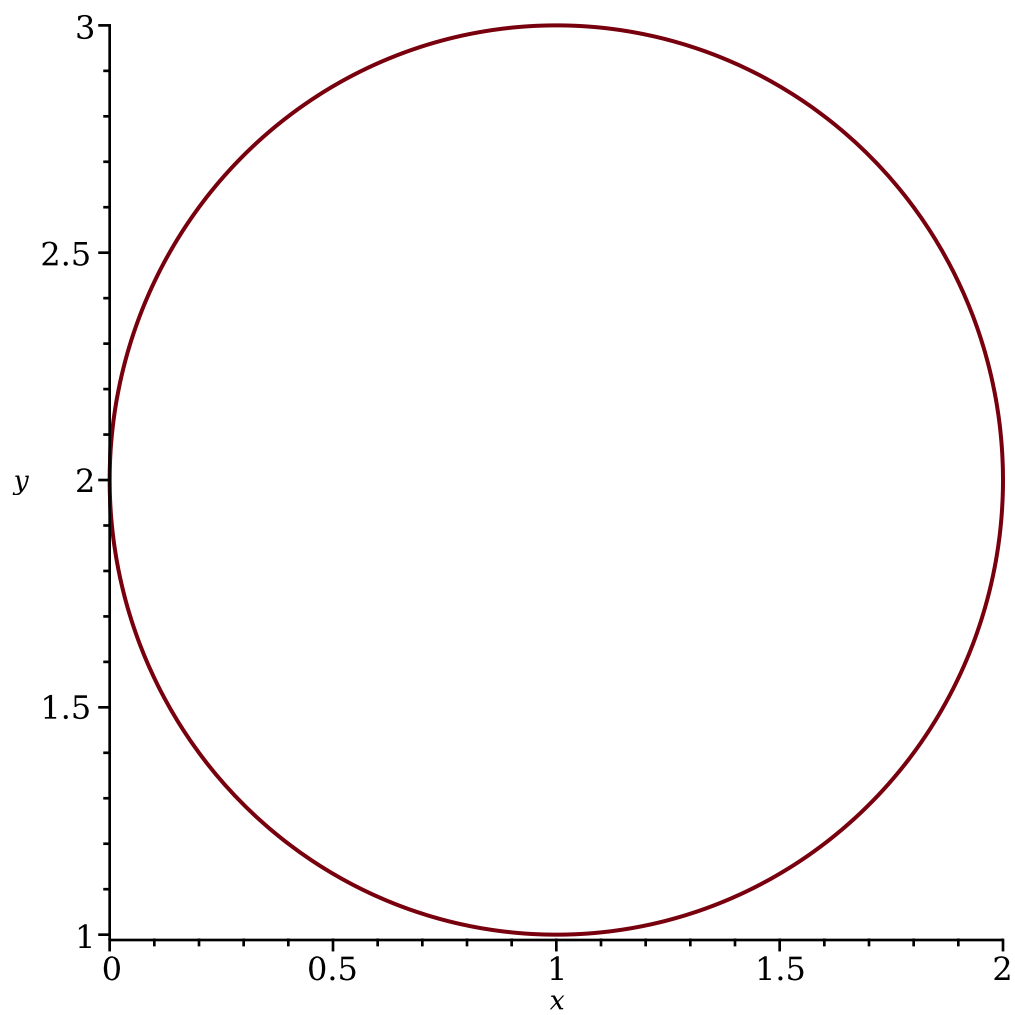
```

> f := (t, s) → 1 - ⎛  $\frac{s \cdot \cos(4t) \cdot \cos(t)}{\sqrt{1 - s^2 \cdot \cos^2(4t) \cdot \sin^2(t)}}$  ⎞ :
> x := (t, s) → f⎛  $t - \frac{\text{Pi}}{2}$ , s ⎞ :
> y := (t, s) → f(t, s) :
> list_xy := ⌈ x⎛  $t, \frac{s}{10}$  ⎞, y⎛  $t, \frac{s}{10}$  ⎞, t = 0..2·Pi ⌋ $ s = 1..10 :
> plot([list_xy])

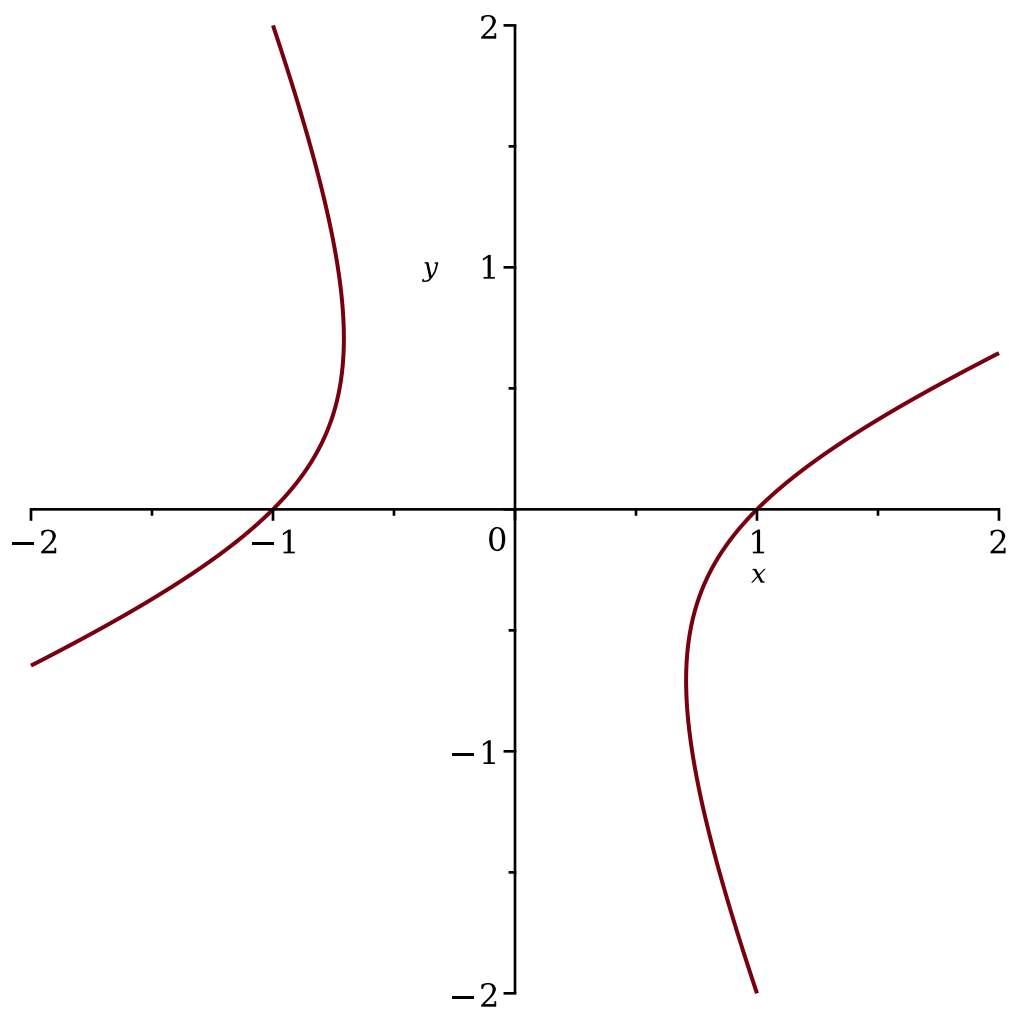
```



```
> with(plots) :
> implicitplot(x^2 + y^2 - 2 x - 4 y + 4 = 0)
```



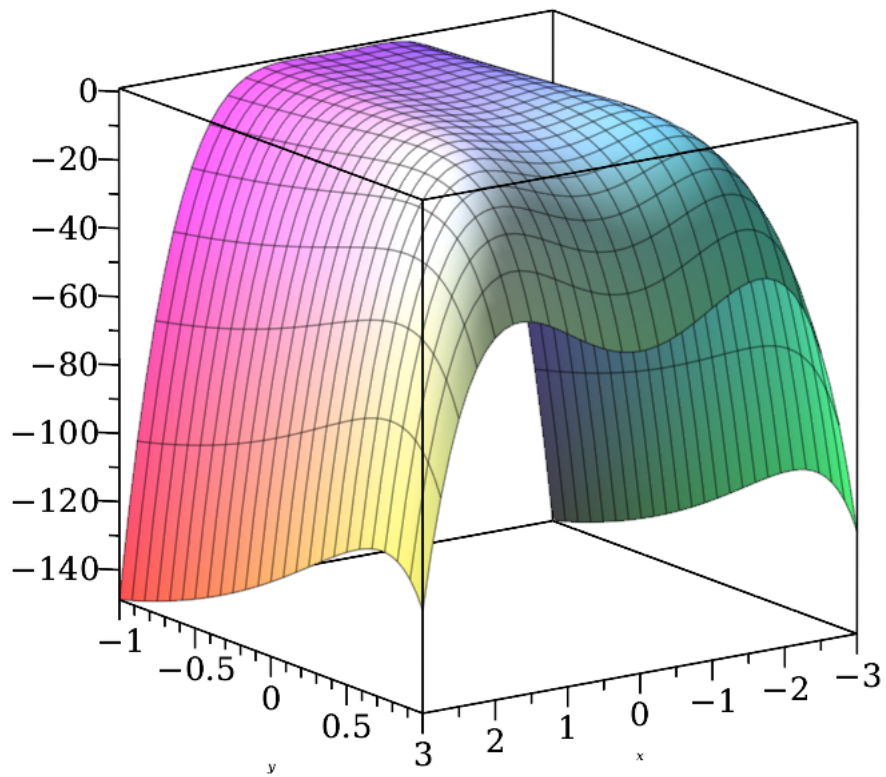
```
> with(plots) :  
> implicitplot( (x - y)^2 - 2 y^2 = 1)
```

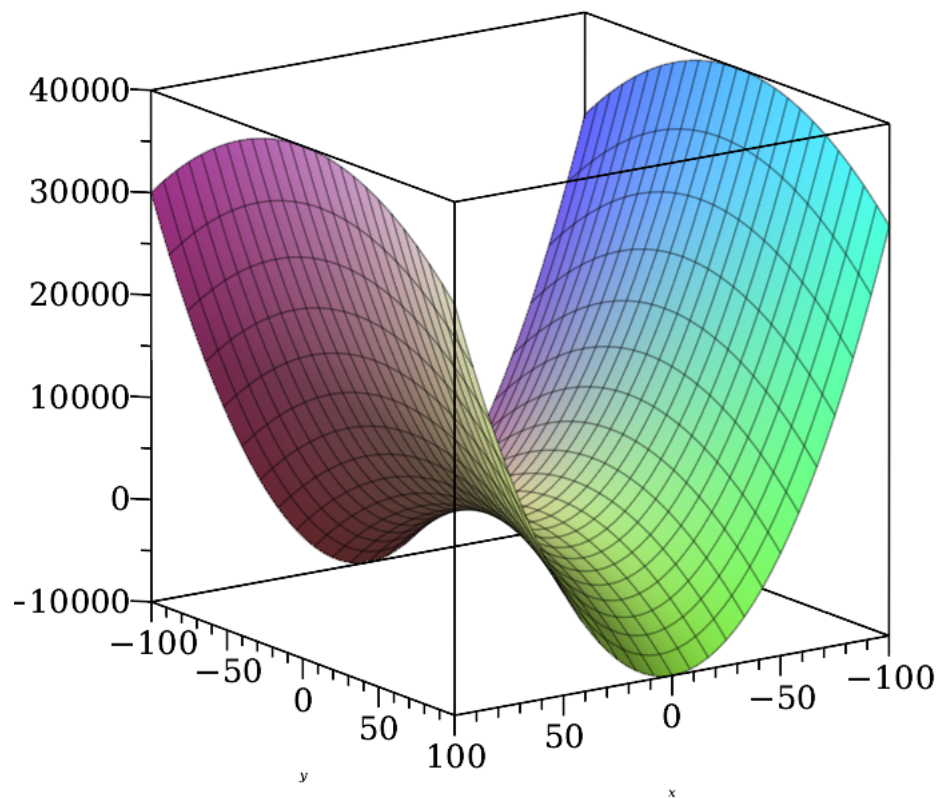
```

> z := (x, y) → 4 x2 · exp(y) − 2 x4 − exp(4 y) :
> with(plots) :
> plot3d(z(x, y), x = −3..3, y = −1..1, axes = boxed)

```



```
> z := (x, y) -> 4 x^2 - y^2 :
> with(plots) :
> plot3d(z(x, y), x = -100..100, y = -100..100, axes = boxed)
```



```

> with(linalg) :
> A := matrix([[1, 2, -1], [0, 1, 0], [3, -1, 2]]) :
> B := matrix([[1, 2, 3], [1, 1, 2], [2, 1, 1]]) :
> C := matrix([[2, 1, 1], [0, 1, -1], [4, 2, 2]]) :
> evalm(2 · A - B&*C);

```

$$\begin{bmatrix} -12 & -5 & -7 \\ -10 & -4 & -4 \\ -2 & -7 & 1 \end{bmatrix}$$

(17)

```

> evalm(A(-1));

```

(18)

$$\begin{bmatrix} \frac{2}{5} & -\frac{3}{5} & \frac{1}{5} \\ 0 & 1 & 0 \\ -\frac{3}{5} & \frac{7}{5} & \frac{1}{5} \end{bmatrix} \quad (18)$$

```
> eigenvects(C);
[0, 1, {[ -1  1  1 ]}], [2, 1, {[ 1  -2  2 ]}], [3, 1, {[ -1  1  -2 ]}] (19)
```

```
> eigenvals(C)
0, 3, 2 (20)
```

```
>
```